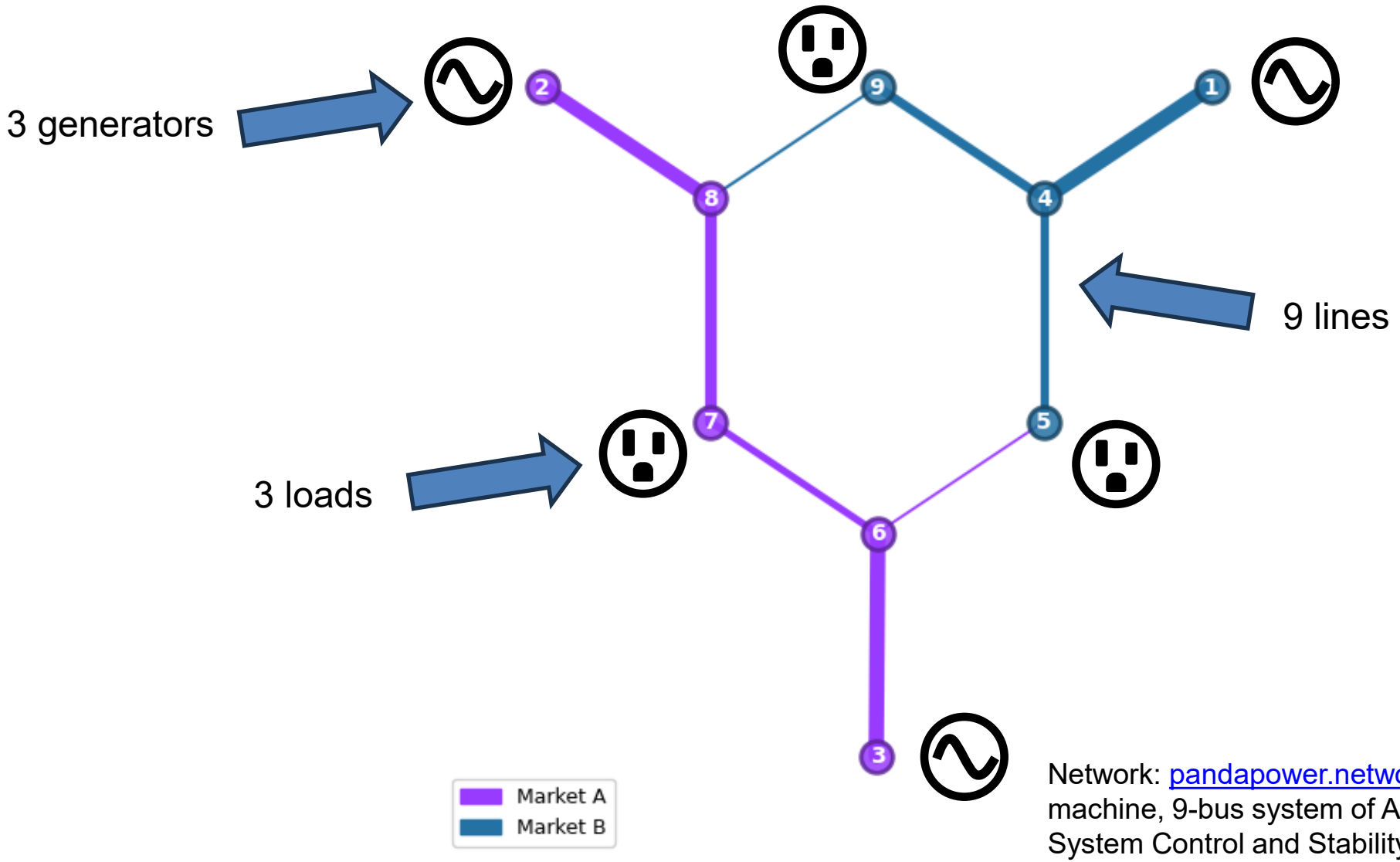
Seams Workshop – Congestion Management Examples

Paul Dockery
Market Strategy & Governance Manager

Congestion management examples using an illustrative model

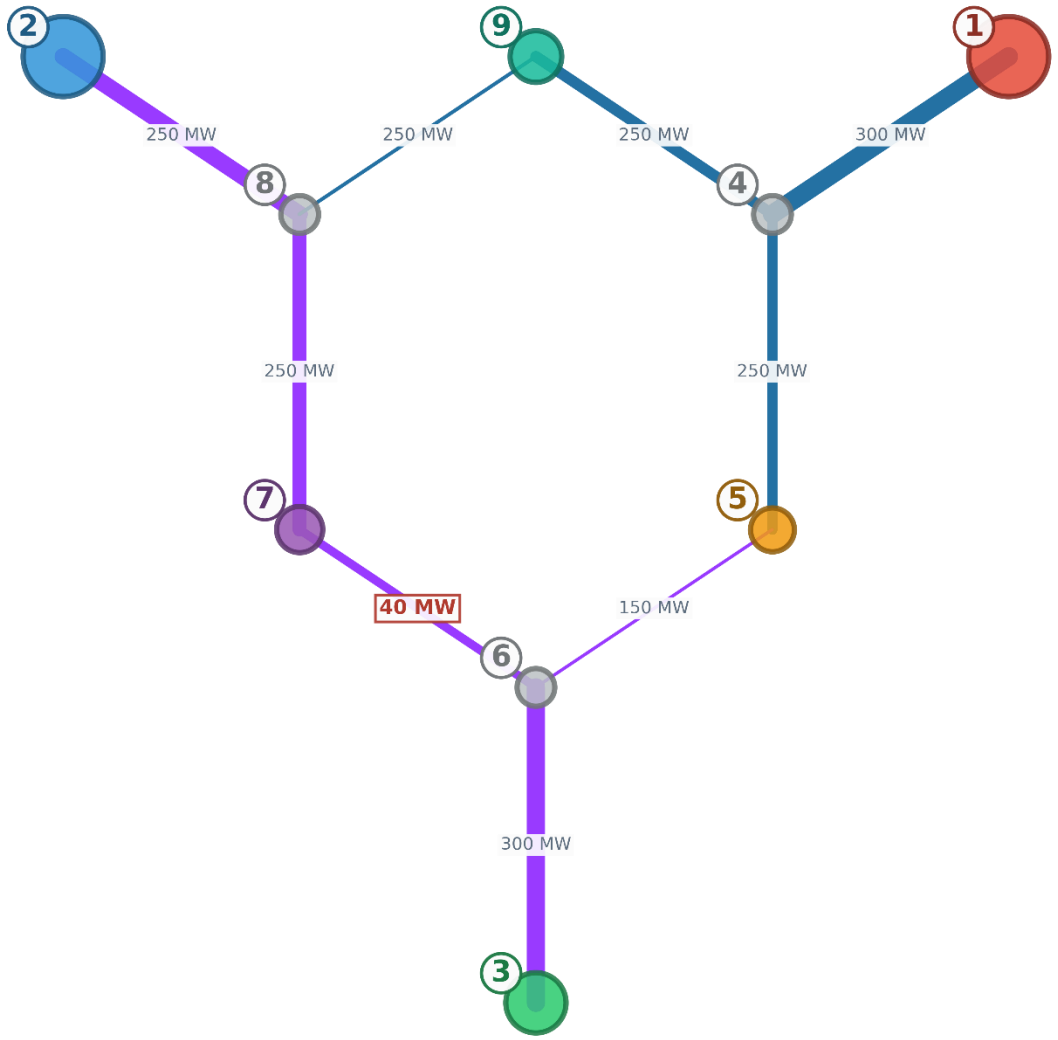
- Forecasts for nodes outside a market footprint
- Dispatch on an illustrative network with a single binding constraint under two examples
  - Ignoring flow contribution from outside the market area
  - Over-forecasting flow contribution from outside the market area
- Production cost and dispatch changes

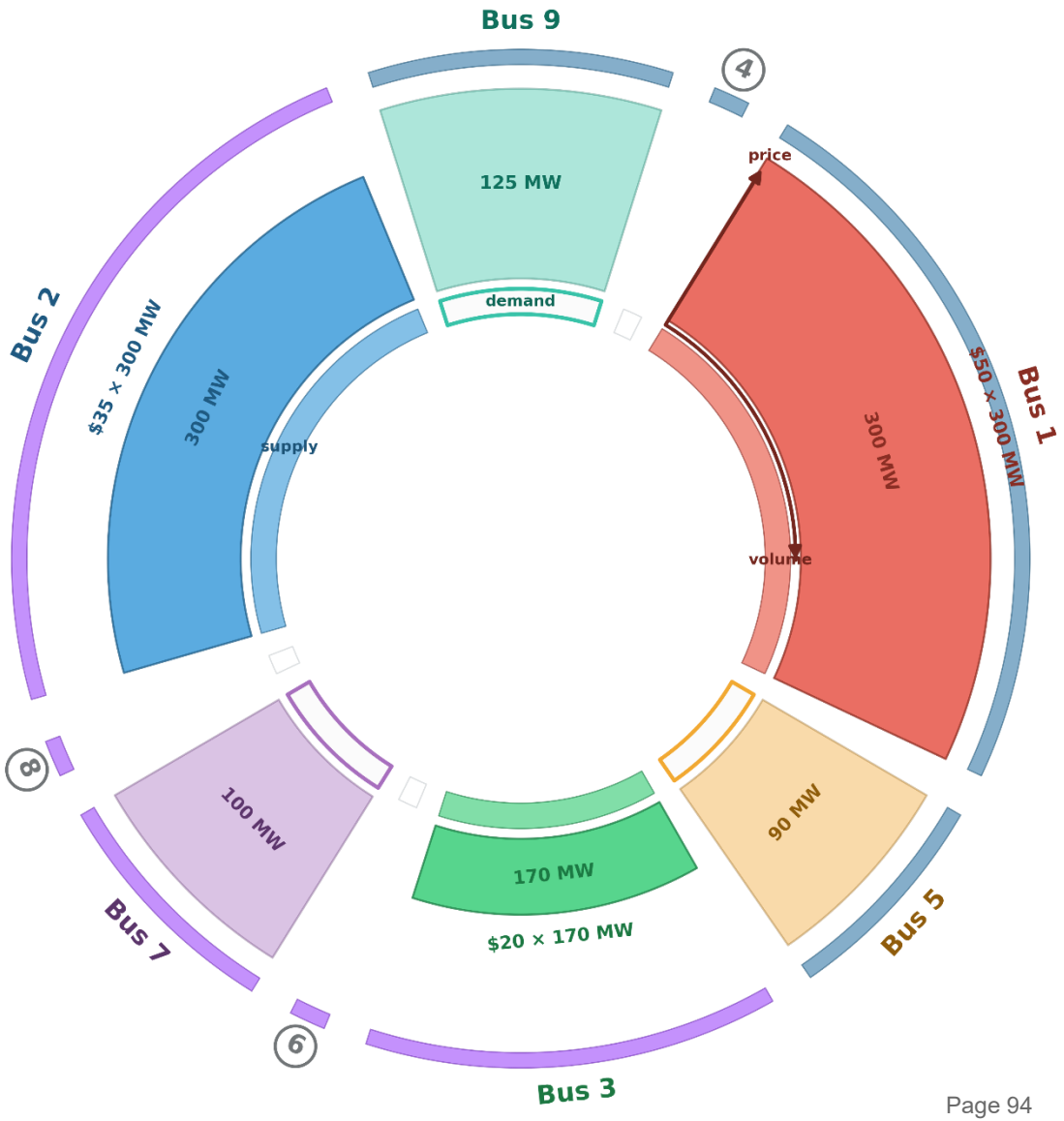
WSCC 9 Bus Test Case – Seams Example



Network: pandapower.networks.case9 after the WSCC 3-machine, 9-bus system of Anderson & Fouad, Power System Control and Stability (1977).

WSCC 9 Bus Test Case – Seams Example

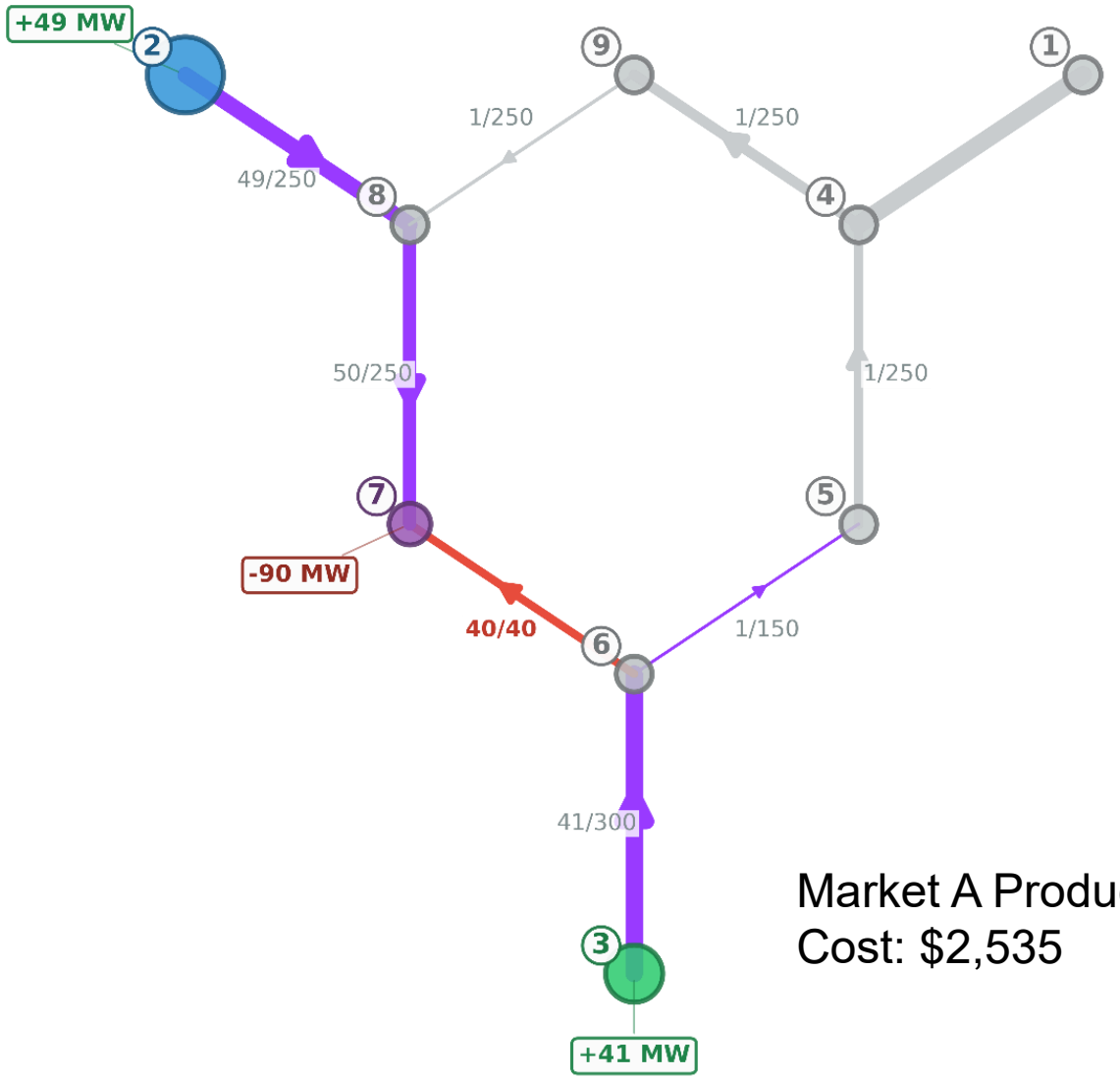




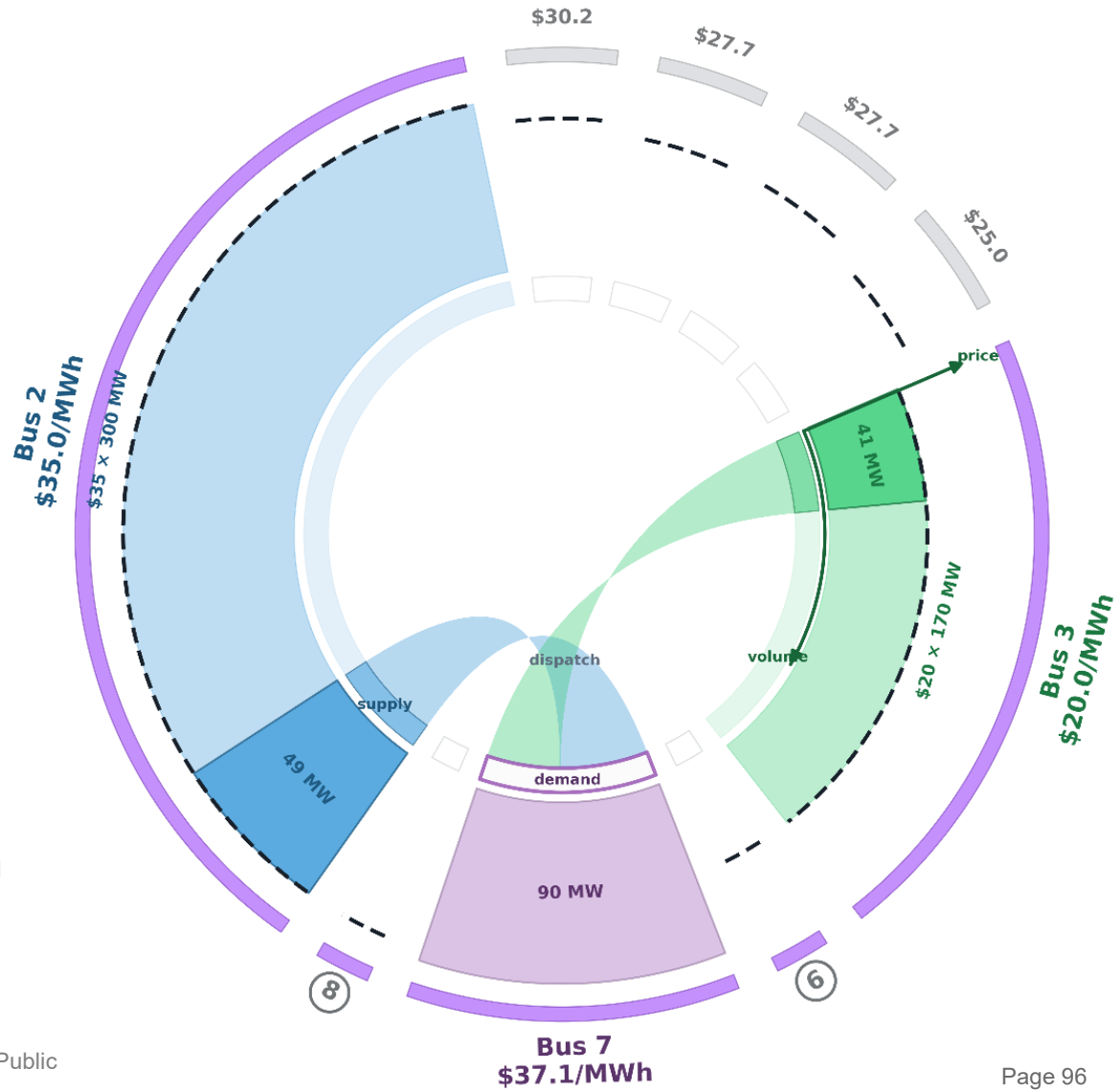
Example #1 – no forecast of neighbor's flow contribution

- Each market clears independently without accounting for each other's use of the transmission network
- Market A has a binding constraint and finds the least cost dispatch of its system resources while respecting the limit
- Market B has no binding constraints and does not monitor binding constraint in Market A
- Power flow on the network based on combined dispatch overloads a line
- Market A redispatches to bring the line back into its limits

Example #1 – no forecast of neighbor's flow contribution – Market A



Market A Production
Cost: $2,535



Example #1 – no forecast of neighbor's flow contribution – Market A

Marginal cost of serving load at bus 7

+ Bus 2 Gen

↑ 1.14 MW @ $35/MWh

= $39.90

- Bus 3 Gen

↓ 0.14 MW @ $20/MWh

= $2.80

△ Bus 7: $39.90 - $2.80 = $37.10



Example #1 – no forecast of neighbor's flow contribution – Market A

Marginal cost of flowing on the line from 6 to 7

+ Bus 2 Gen

↑ 1.34 MW @ $35/MWh

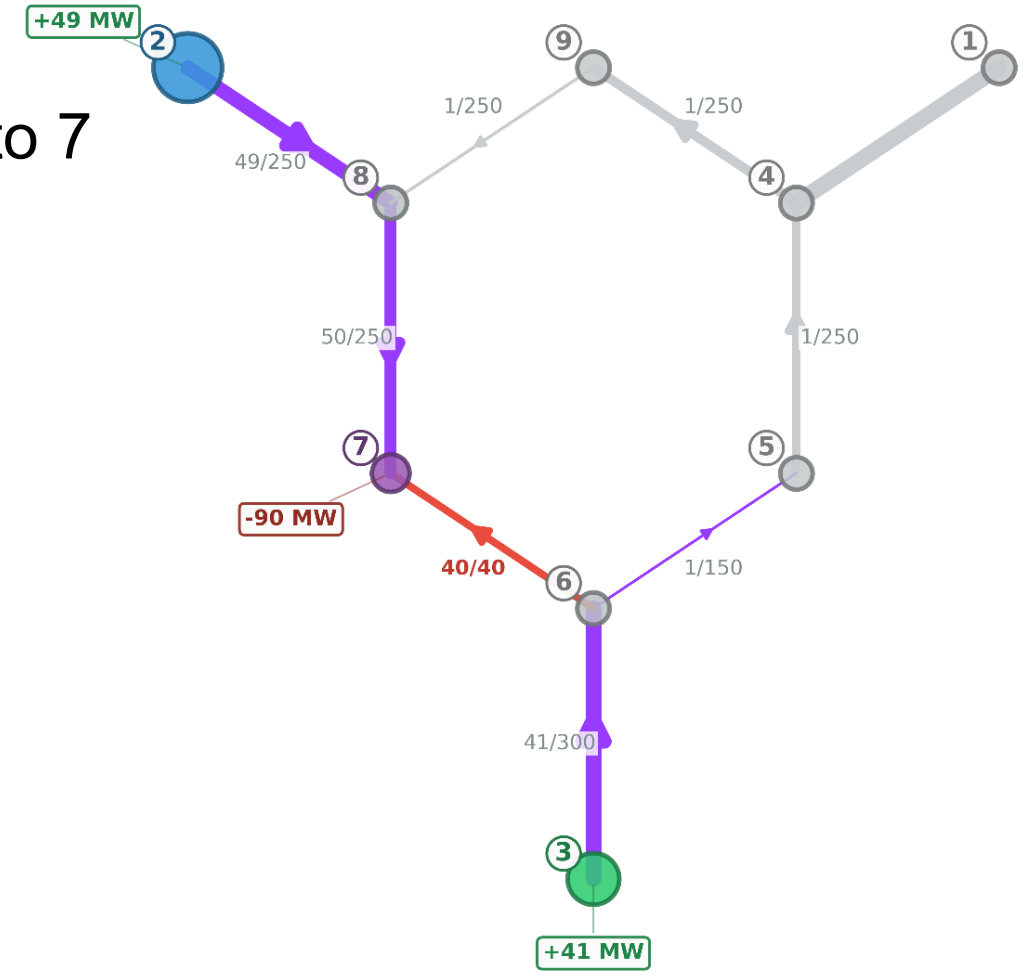
= $46.90

- Bus 3 Gen

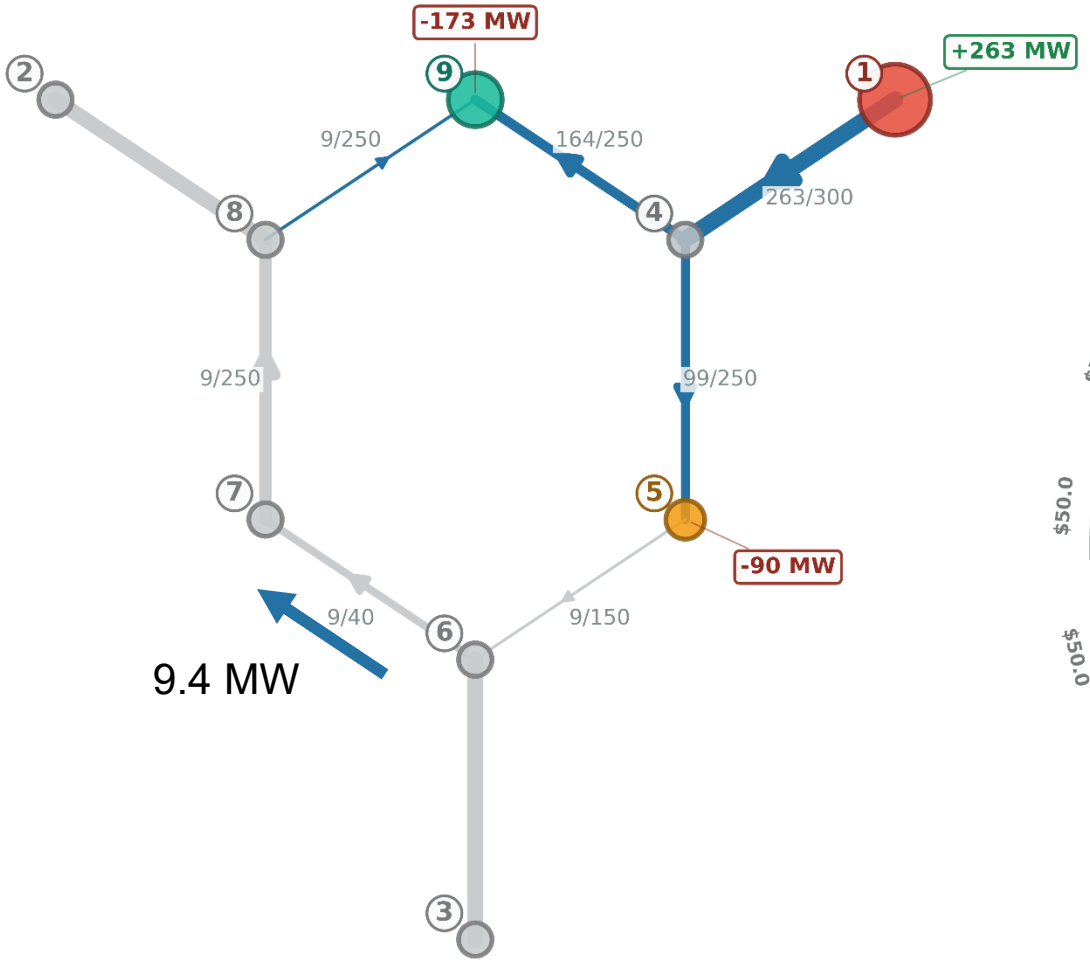
↓ 1.34 MW @ $20/MWh

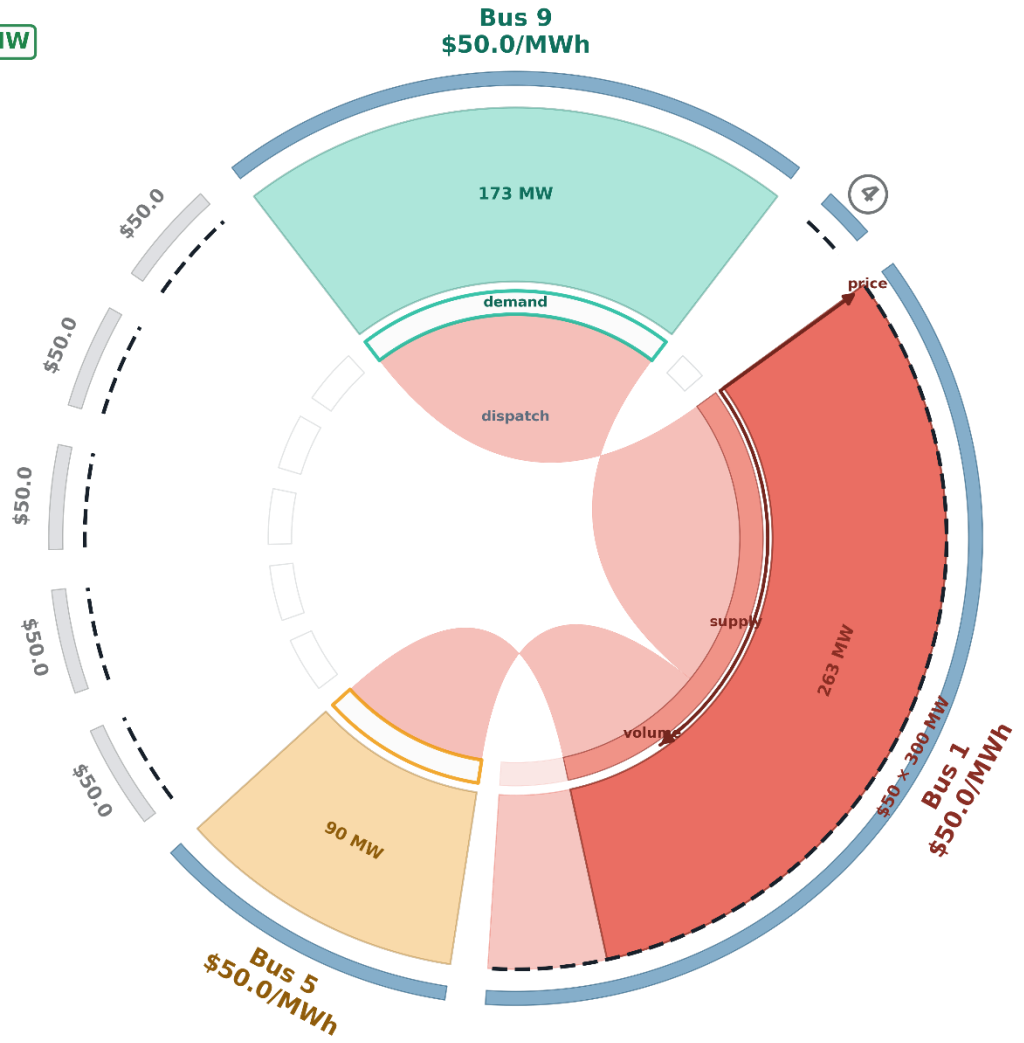
= $26.80

△ Line 6-7: $46.90 - $26.80 = $20.10

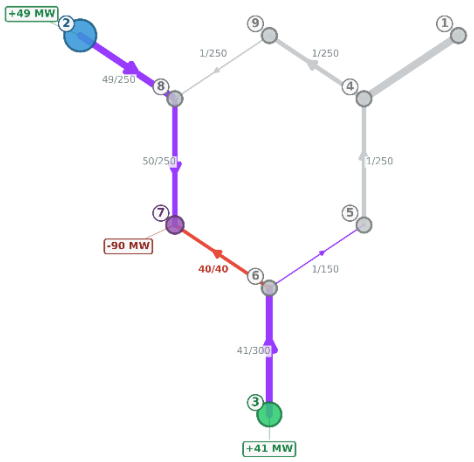


Example #1 – no forecast of neighbor's flow contribution – Market B

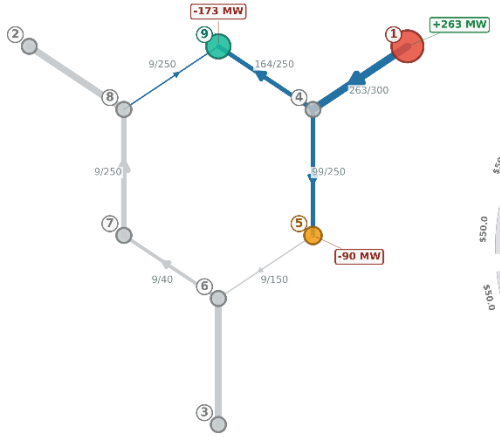


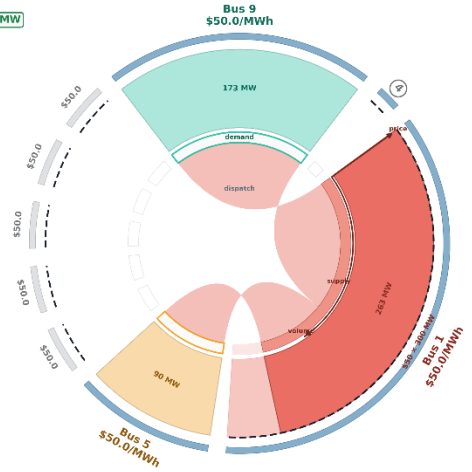


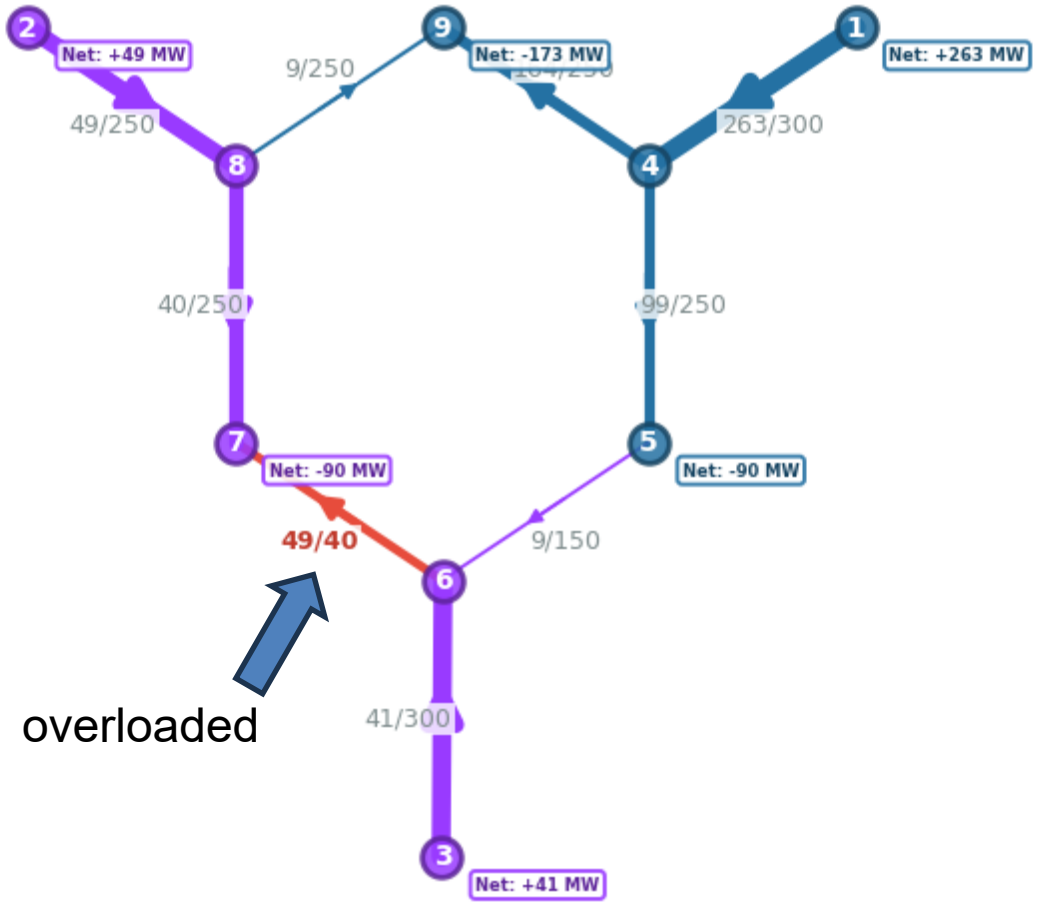
Example #1 – no forecast of neighbor's flow contribution - combined











Example #1 – no forecast of neighbor's flow contribution – redispatch

Market A Production Cost: $2,725

Independent run: $2,535

+ Bus 2 Gen

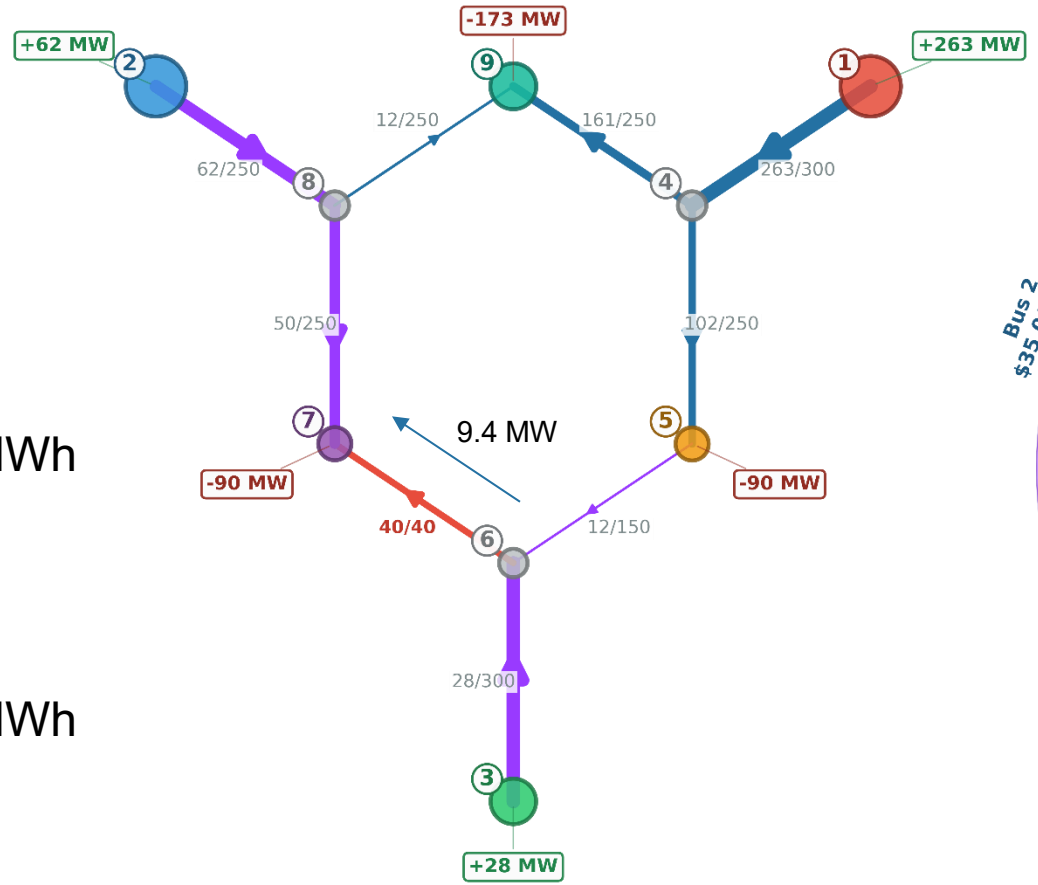
↑ 12.66 MW @ $35/MWh

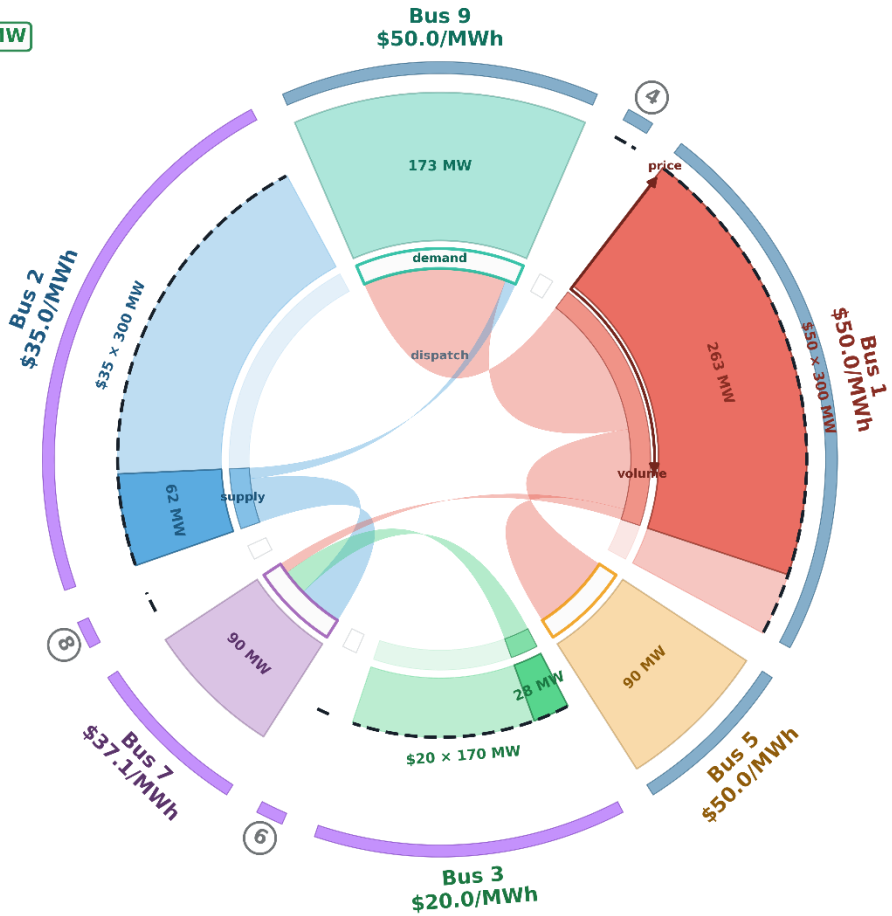
= $443

- Bus 3 Gen

↓ 12.66 MW @ $20/MWh

= $253





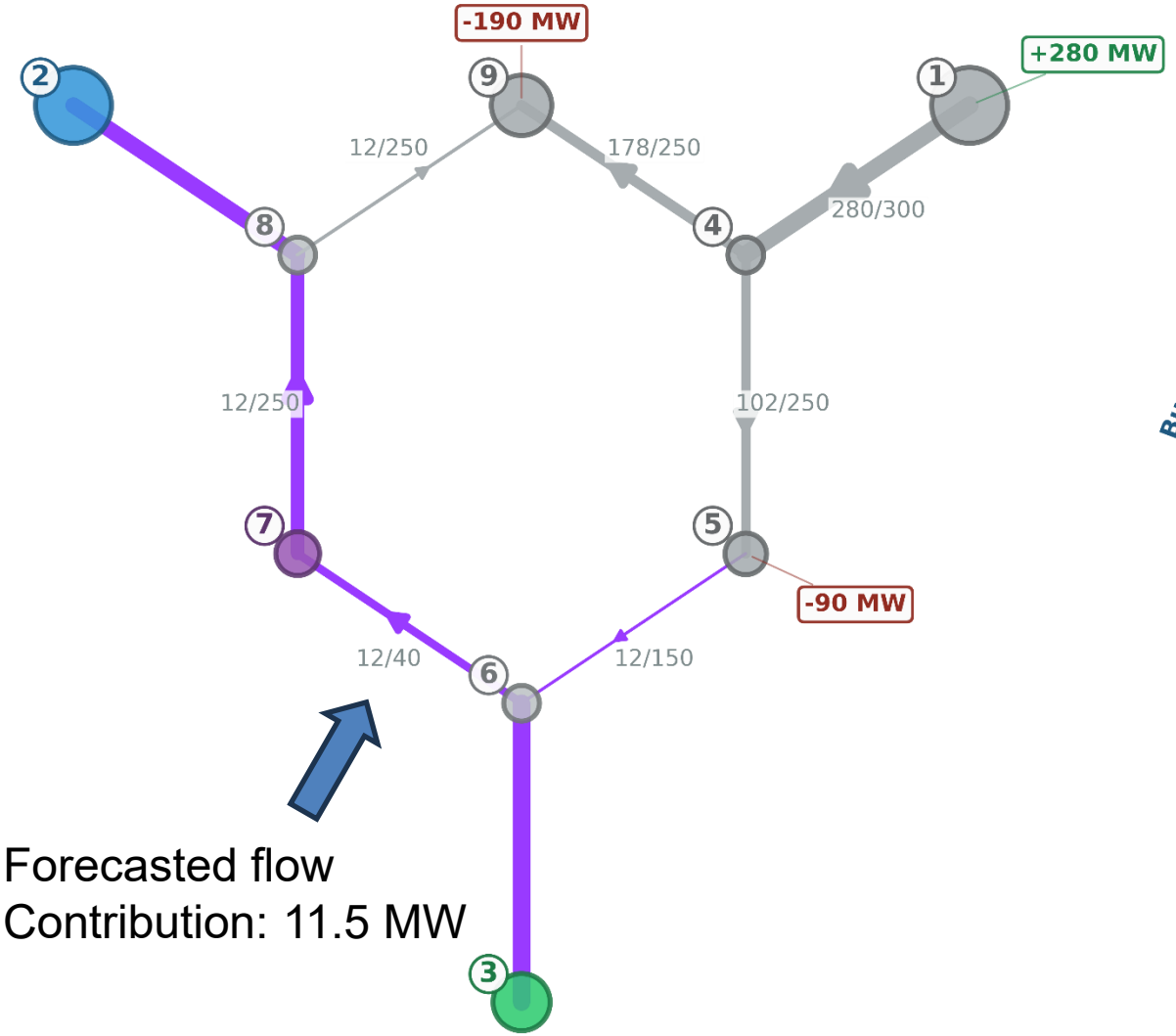
△ Market A production cost: $443 - $253 = $190

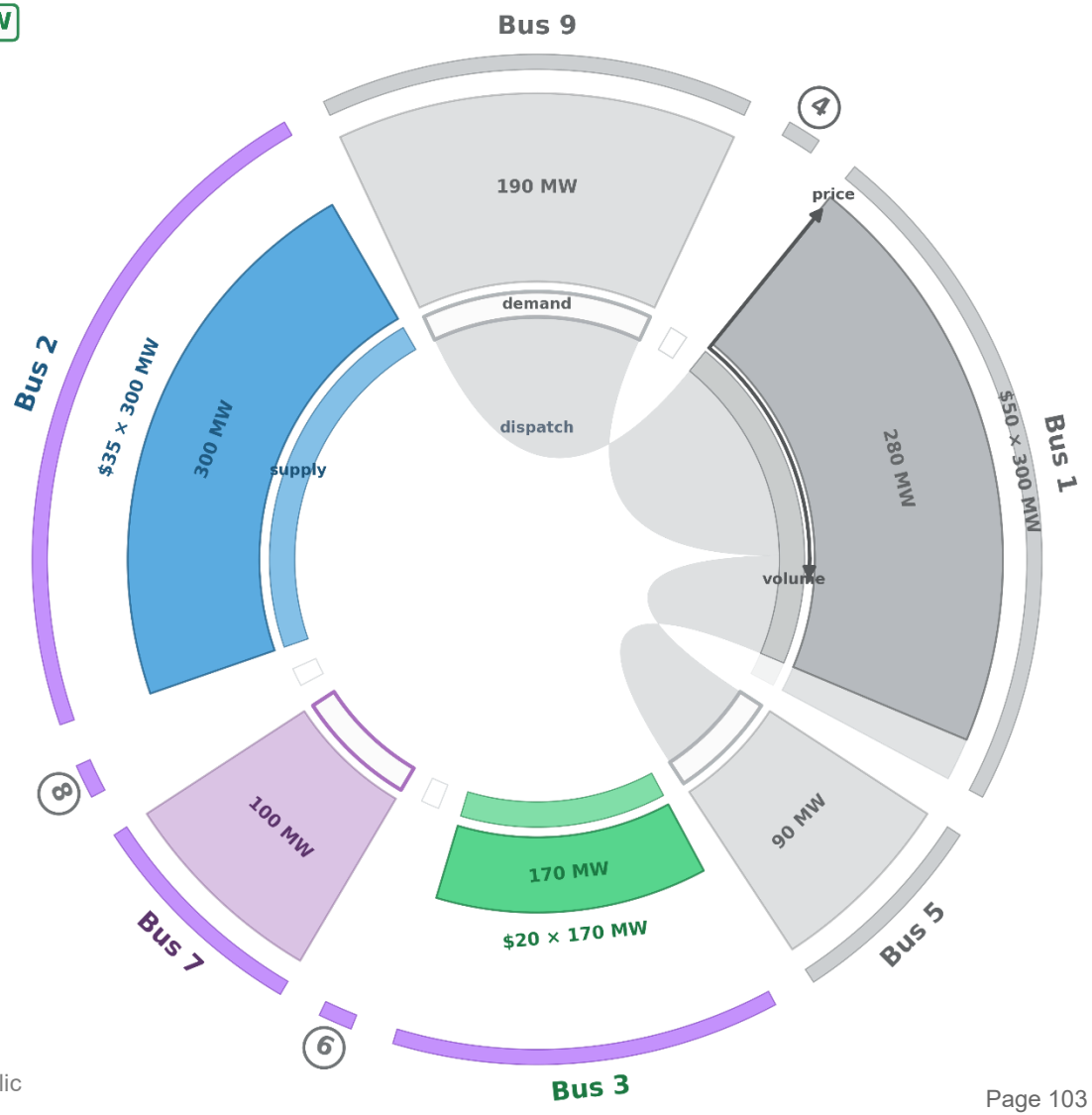
= 9.4 MW @ $20.10/MWh = $190

Example #2 – over-forecast neighbor's flow contribution

- Each market forecasts the other's use of the transmission network
- Market A has a binding constraint and finds the least cost dispatch of its system resources while respecting the limit and accounting for Market B's expected flows
- Market B has no binding constraints and does not monitor binding constraint in Market A
- Power flow on the network based on combined dispatch leaves space on the line
- Market A bears the cost of phantom congestion

Example #2 – over-forecast neighbor's flow contribution – Market A input





Example #2 – over-forecast neighbor's flow contribution – Market A

Market A Production Cost: $2,767

Independent run: $2,535

+ Bus 2 Gen

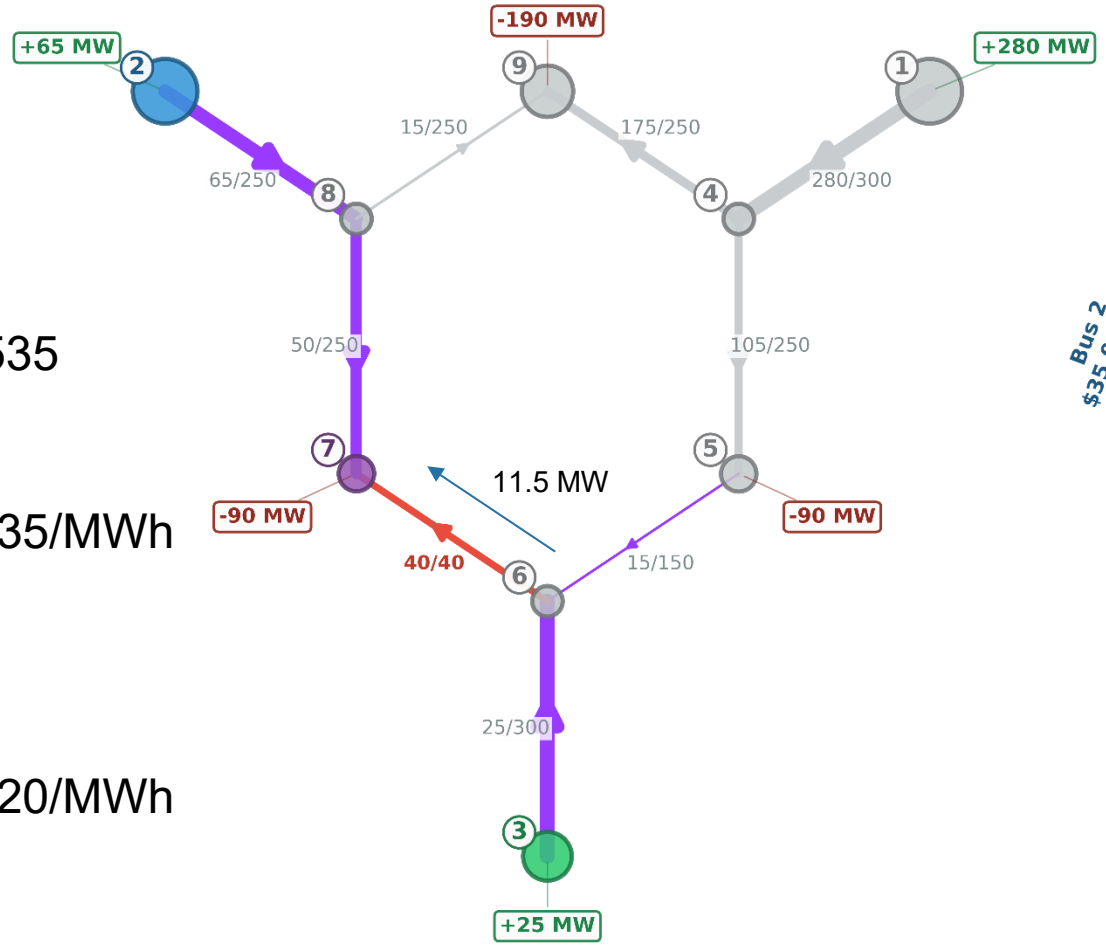
↑ 15.45 MW @ $35/MWh

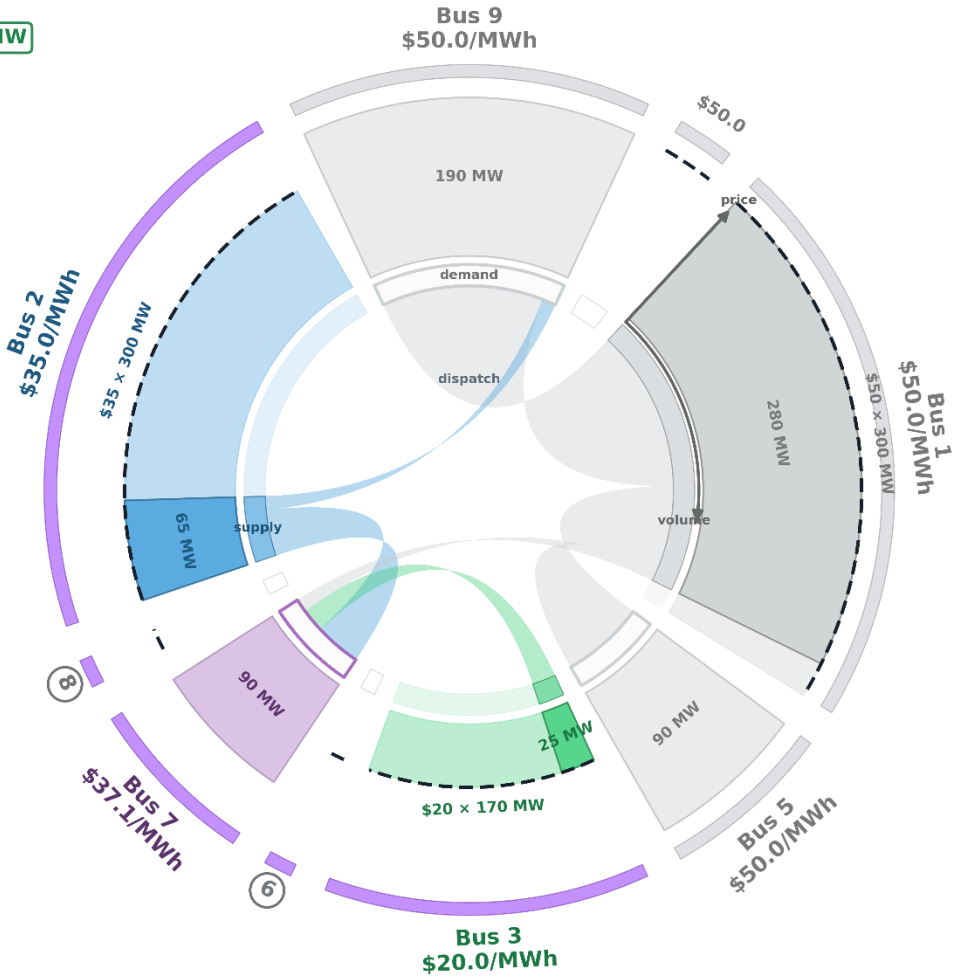
= $541

- Bus 3 Gen

↓ 15.45 MW @ $20/MWh

= $309

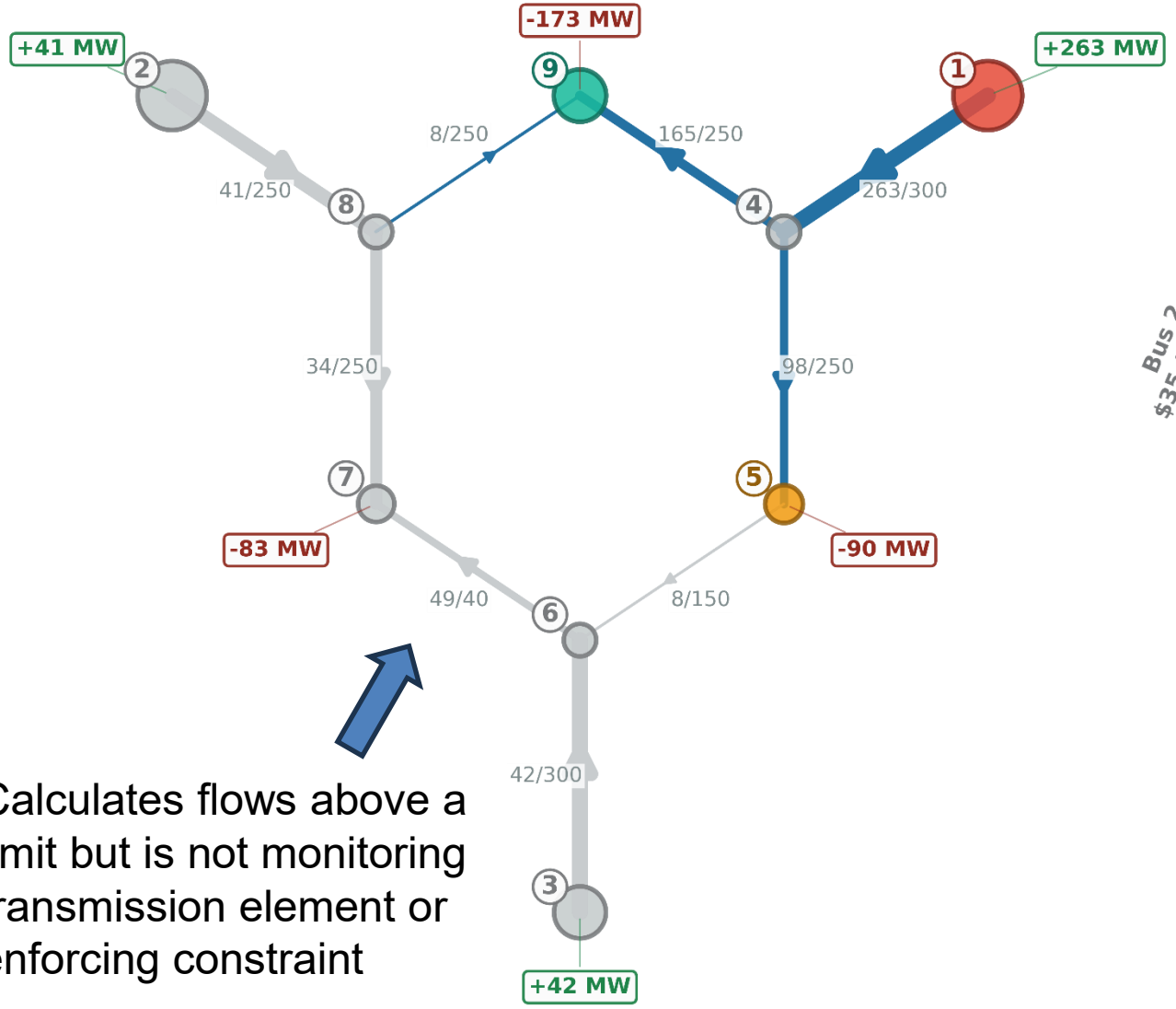




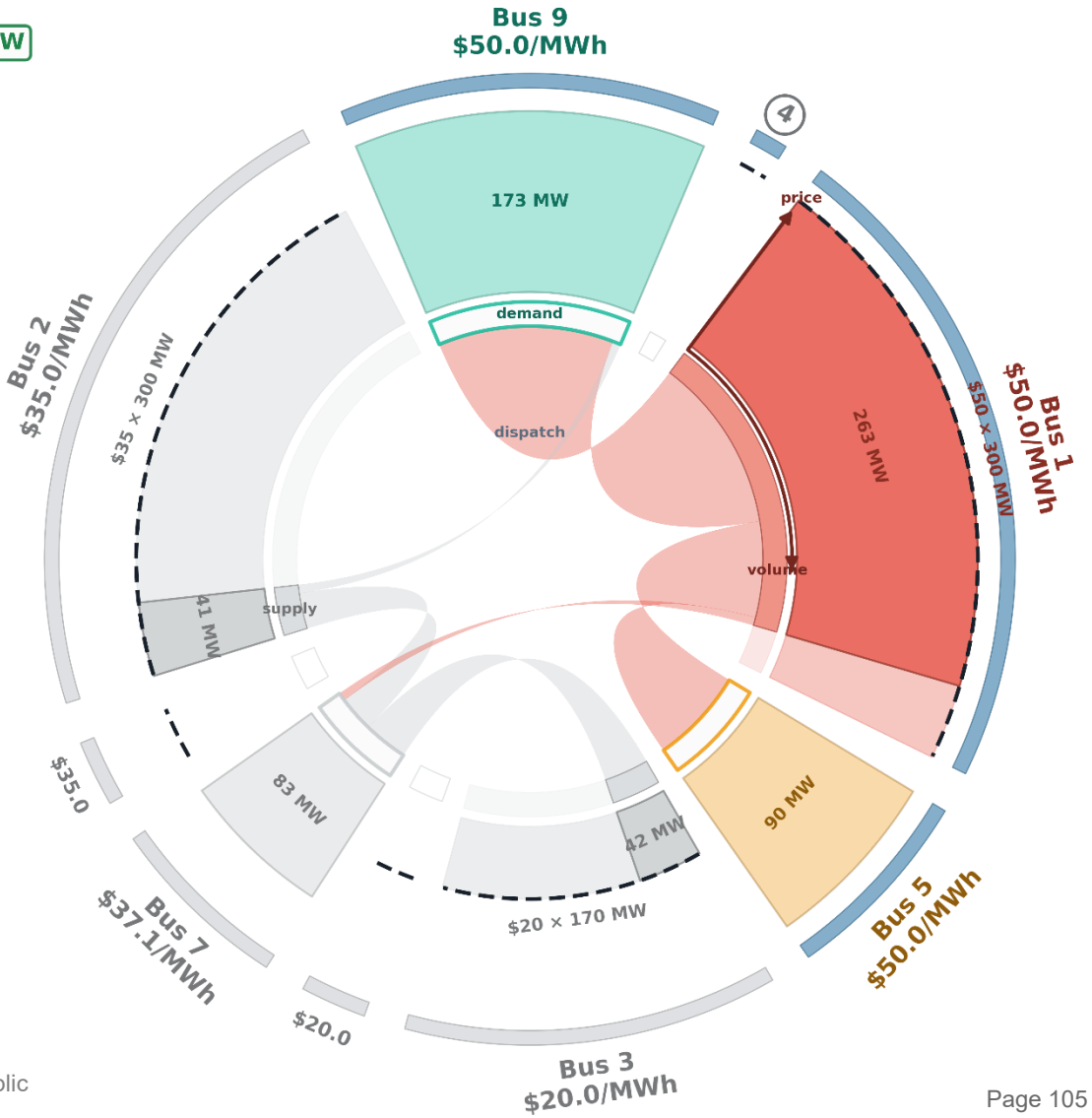
△ Market A production cost: $541 - $309 = $232

= 11.5 MW @ $20.10/MWh = $232

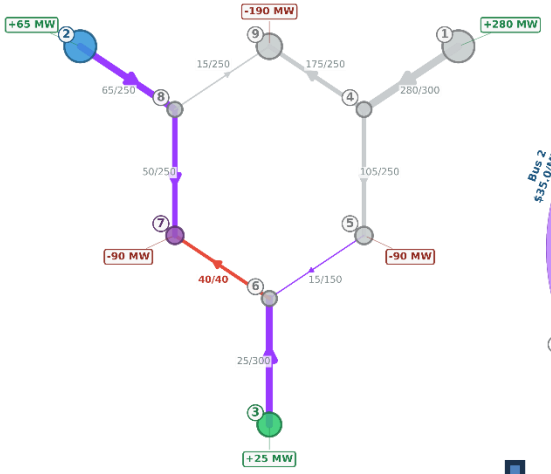
Example #2 – over-forecast neighbor's flow contribution – Market B

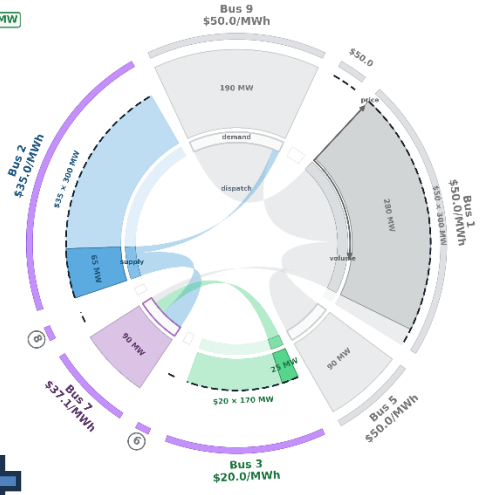


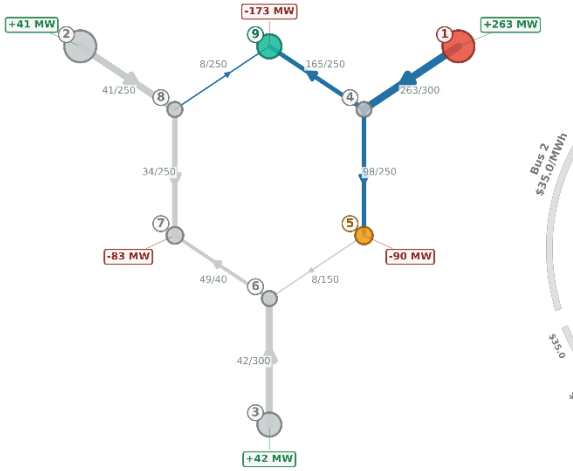
Calculates flows above a limit but is not monitoring transmission element or enforcing constraint

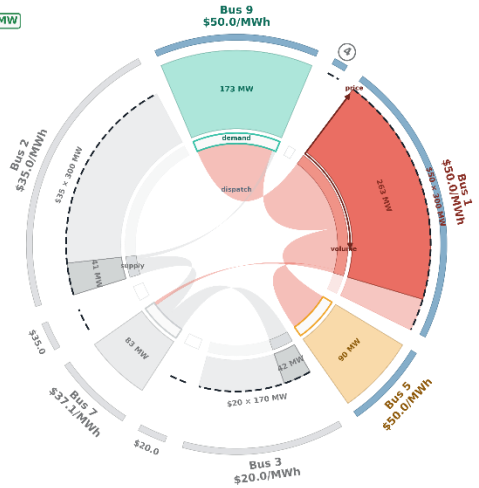


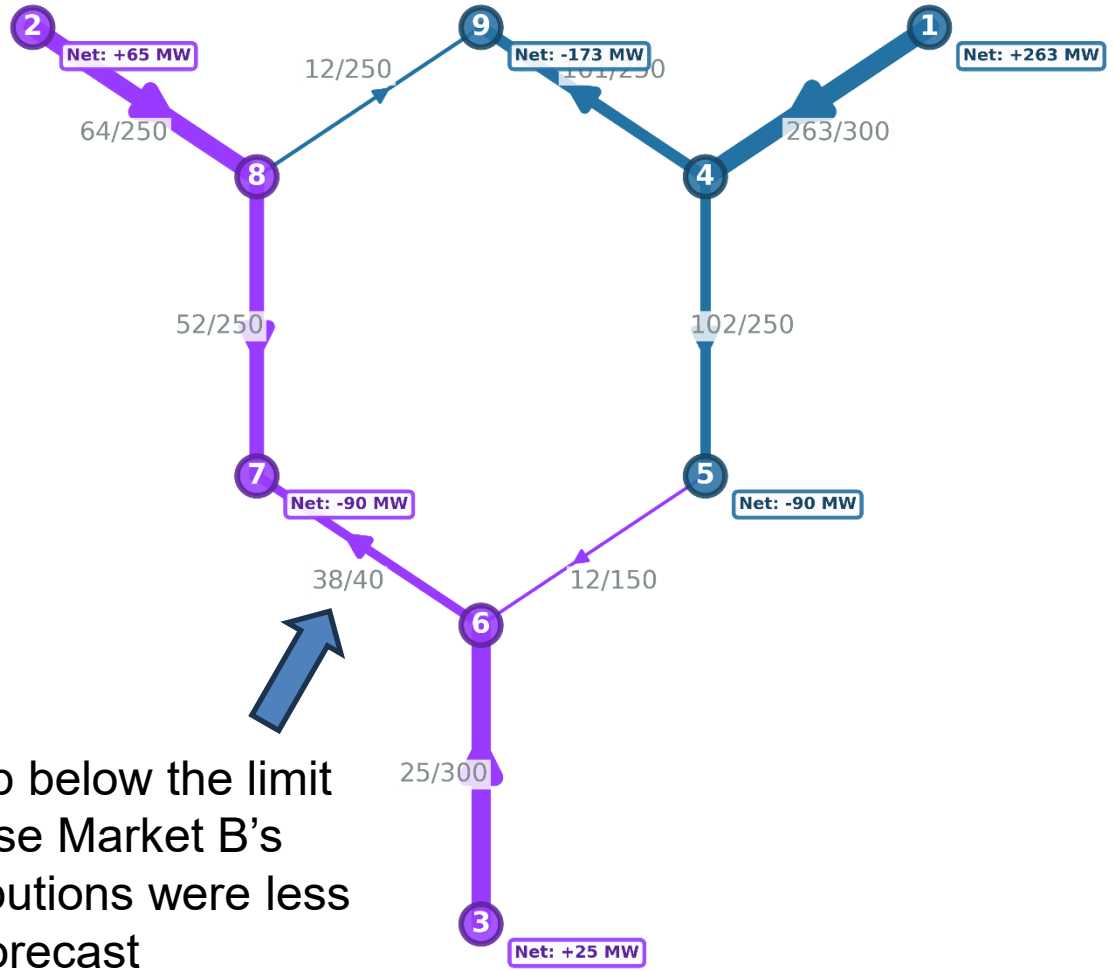
Example #2 – over-forecast neighbor's flow contribution





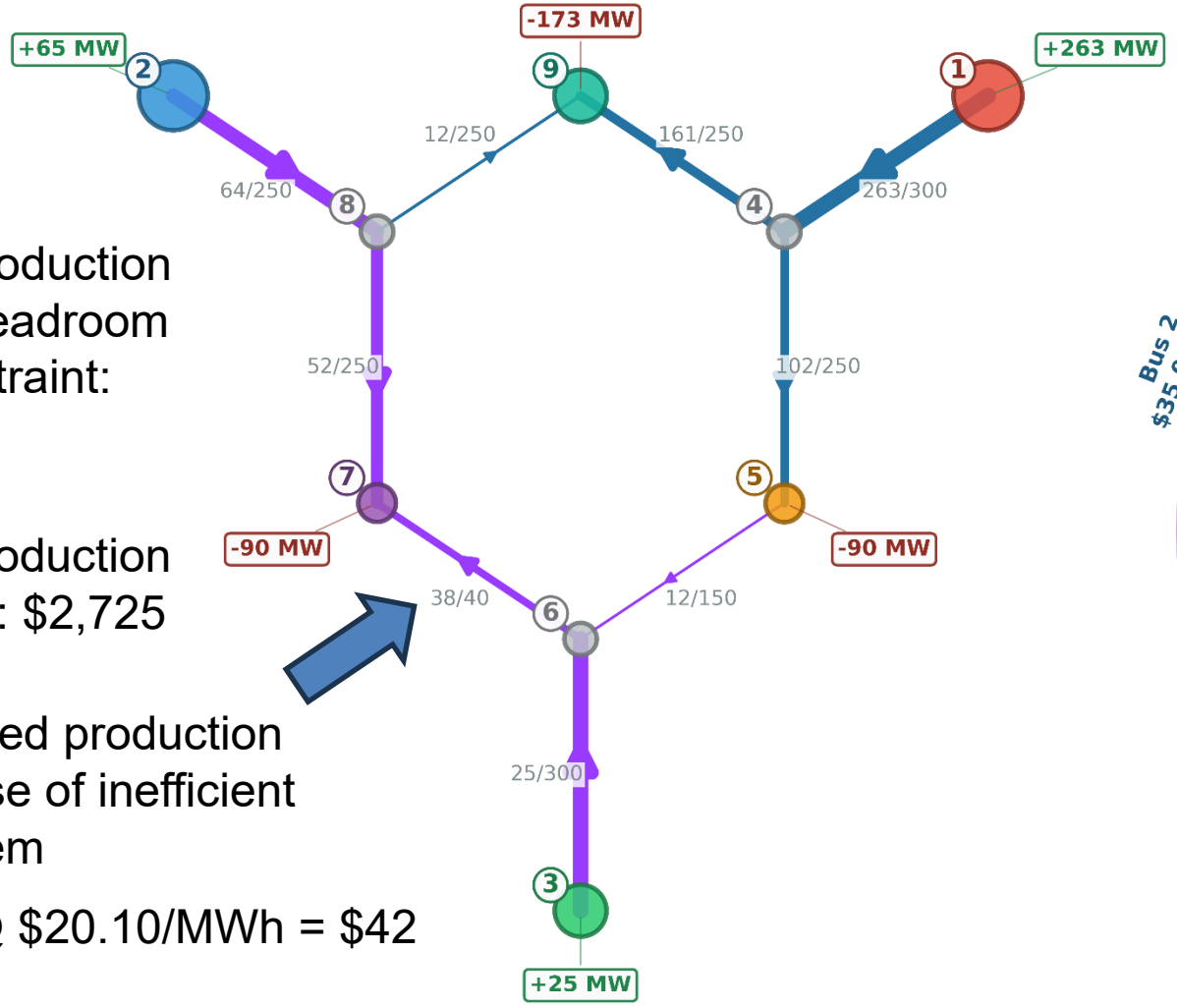






End up below the limit because Market B's contributions were less than forecast

Example #2 – over-forecast neighbor's flow contribution

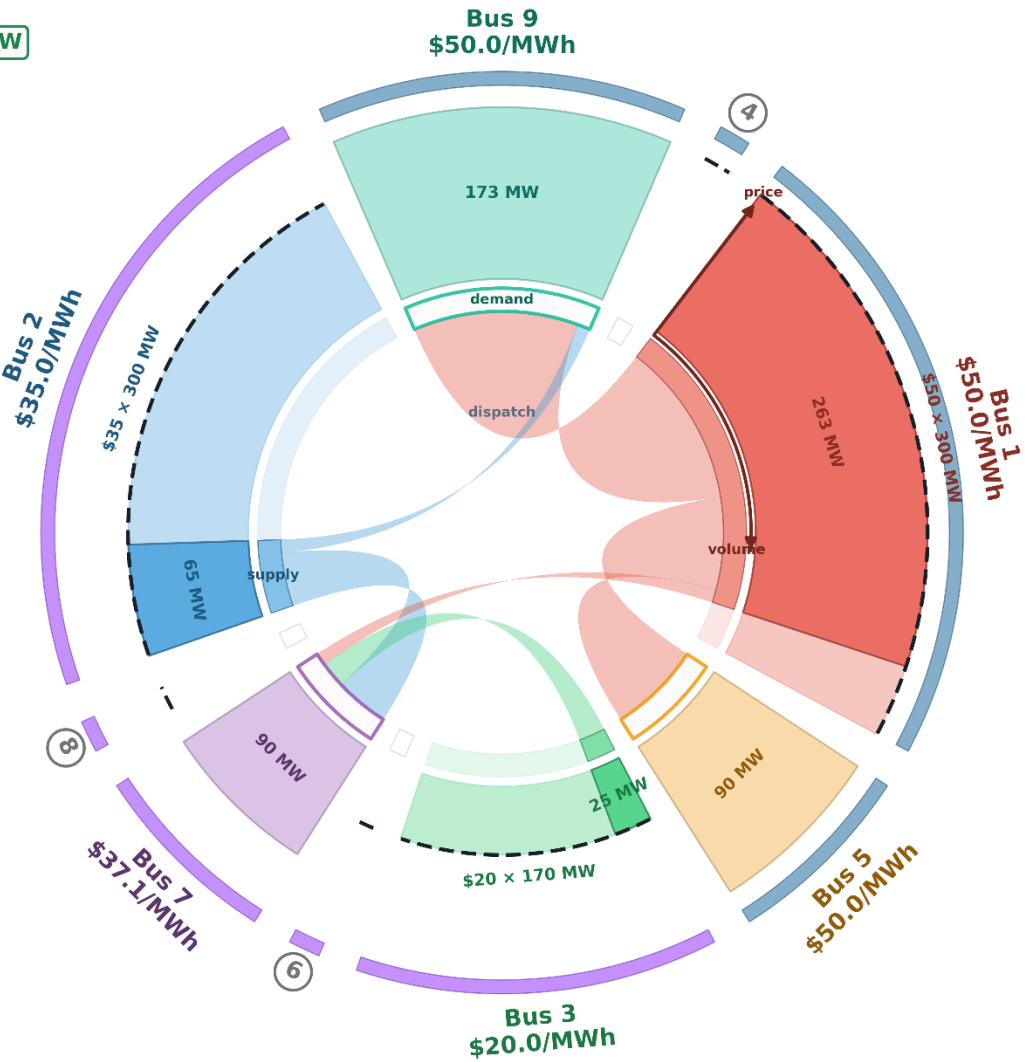


Market A Production Cost with headroom on the constraint: $2,767

Market A Production Cost at limit: $2,725

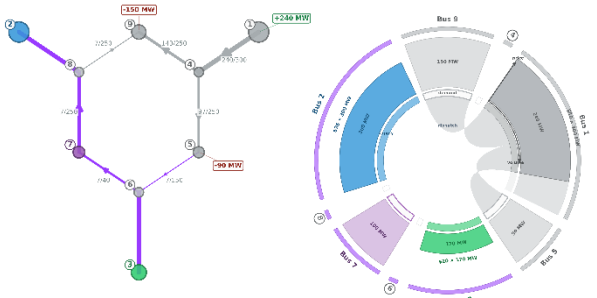
$42 increased production cost because of inefficient use of system

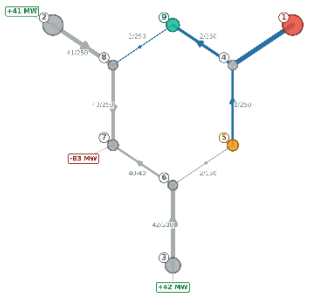
= 2.1 MW @ $20.10/MWh = $42

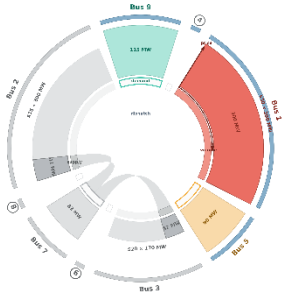


Parallel process considerations

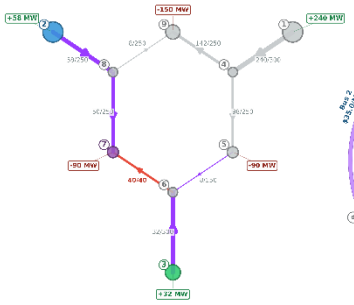
Inputs

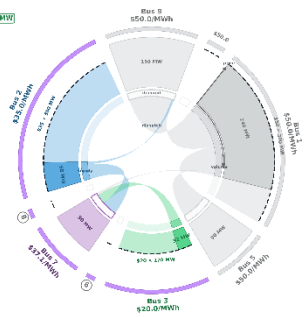


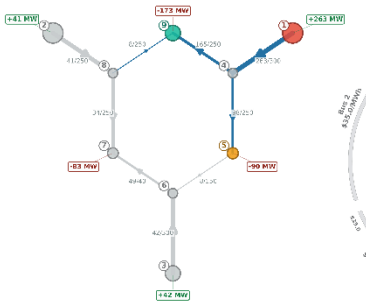


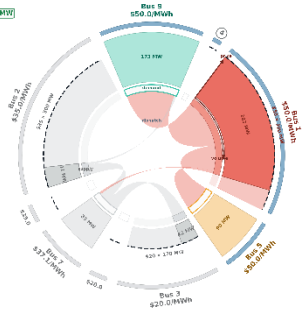


Optimization

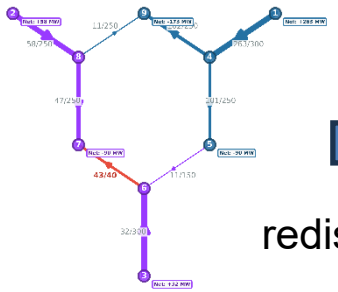




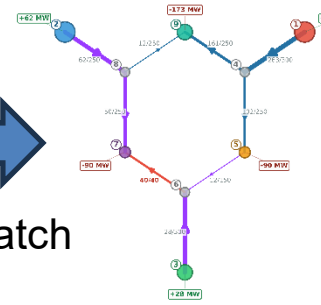


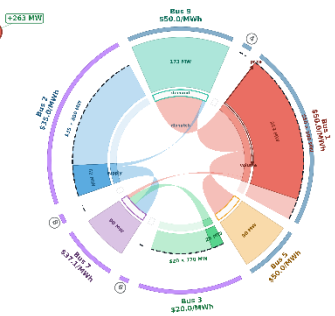


Reconciliation



redispatch





Key considerations

- Data sharing
- Modeling
- Methodology for managing jointly impacted constraints